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3. AIR QUALITY IMPACT

3.1 Introduction

3.1.1 This section presents the assessment on potential air quality impacts on the air sensitive receivers (ASRs) arising from construction and operation of the Project. Assessment has been conducted in accordance with the criteria and guidelines as stipulated in Annex 4 and Annex 12 of the technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) as well as the requirements given in Clause 3.5.3 of the EIA Study Brief (No. ESB-375/2025).

3.2 Environmental Legislation, Standards and Guidelines

3.2.1 The Air Pollution Control Ordinance provides the statutory authority for controlling air pollutants from a variety of sources. The Hong Kong Air Quality Objectives (AQOs), which stipulate the maximum allowable concentrations over specific periods for typical pollutants, should be met. The prevailing AQOs are implemented in April 2025 have been adopted as the assessment criteria, as shown in **Table 3.1.**

Table 3.1 Hong Kong Air Quality Objectives

Pollutants	Averaging Time	Concentration Limit (μg/m^3) [1]	Number of Exceedance Allowed per Year
Respirable Suspended Particulates (RSP or PM_{10}) [2]	24-hour	75	9
	Annual	30	Not applicable
Fine Suspended Particulates (FSP or $PM_{2.5}$) [3]	24-hour	37.5	18
	Annual	15	Not applicable
Nitrogen Dioxide (NO_2)	1-hour	200	18
	24-hour	120	9
	Annual	40	Not applicable
Sulphur Dioxide (SO_2)	10-min	500	3
	24-hour	40	3
Carbon Monoxide (CO)	1-hour	30,000	0
	8-hour	10,000	0
	24-hour	4,000	0
Ozone	8-hour	160	9
	Peak season	100	Not applicable
Lead (Pb)	Annual	0.5	Not applicable

Notes:
[1] All measurements of the concentration of gaseous air pollutants, i.e., sulphur dioxide, nitrogen dioxide, ozone and carbon monoxide, are to be adjusted to a reference temperature of 293 Kelvin and a reference pressure of 101.325 kilopascal.
[2] Suspended particulates in air with a nominal aerodynamic diameter of 10μm or smaller.
[3] Suspended particulates in air with a nominal aerodynamic diameter of 2.5μm or smaller.

Air Pollution Control (Construction Dust) Regulation

3.2.2 Notifiable and regulatory works are under the control of Air Pollution Control (Construction Dust) Regulation. Notifiable works include site formation, reclamation, demolition, foundation and superstructure construction for buildings and road construction. Regulatory works include building renovation, road opening and resurfacing, slope stabilisation, and other activities including stockpiling, dusty material handling, excavation, concrete production, etc. This Project is expected to include notifiable works (foundation and superstructure construction and demolition) and regulatory works (dusty material handling and excavation). Contractors and site agents are required to inform Environmental Protection Department (EPD) and adopt air quality control measures to minimise air emission, while carrying out construction works, to the acceptable level.

Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation

3.2.3 The Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation came into effect on 1 June 2015. Under the Regulation, Non-road mobile machinery (NRMMs), except those exempted, are required to comply with the prescribed emission standards. From 1 September 2015, all regulated machines sold or leased for use in Hong Kong must be approved or exempted with a proper label in a prescribed format issued by EPD. Starting from 1 December 2015, only approved or exempted NRMMs with a proper label are allowed to be used in specified activities and locations including construction sites. The Contractor is required to ensure the adopted machines or non-road vehicle under the Project could meet the prescribed emission standards and requirement.

Air Pollution Control (Fuel Restriction) (Amendment) Regulation

3.2.4 The Air Pollution Control (Fuel Restriction) (Amendment) Regulation 2024 was to impose legal control on the type of fuels allowed for use and their sulphur contents in commercial and industrial processes to reduce sulphur dioxide (SO_2) emissions. This regulation was commenced on 1 April 2025 and the sulphur content of liquid fuel has been tightened to 0.001% by weight.

Development Bureau Technical Circular (Works)

3.2.5 The *Development Bureau Technical Circular (Works) No. 13/2020 (DEVB TC(W) No.13/2020)* Timely Application of Temporary Electricity and Water Supply for Public Works Contract and Wider Use of Electric Vehicles in Public Works Contracts is one of the environmental guidelines on timely application of temporary electricity and wider use of electric vehicles (EV) in public works contract. The project team should also specify the use of EV(s) and installation of a designated medium-speed charger for each EV in each public contract.

3.2.6 The *Development Bureau Technical Circular (Works) No. 1/2015* Emissions Control of Non-Road Mobile Machinery in Capital Works Contracts of Public Works also requires that no exempted generators, air compressors, excavators and crawler cranes shall be allowed in the new capital works contracts of public works (including design and build contracts) with an estimated contract value exceeding $200 million, unless is at the discretion of the Architect/Engineer considering no feasible alternative.

Practice Note on Control of Air Pollution in Vehicle Tunnels

3.2.7 The Practice Note on Control of Air Pollution in Vehicle Tunnels prepared by EPD provides guidelines on control of air pollution in vehicle tunnels. Guideline values on tunnel air quality are presented in **Table 3.2**.

Table 3.2 Tunnel Air Quality Guidelines

Pollutant	Averaging Time	Maximum Concentration	
		µg/m^3 [1]	ppm
Carbon Monoxide (CO)	5 minutes	115,000	100
Nitrogen Dioxide (NO_2)	5 minutes	1,800	1
Sulphur Dioxide	5 minutes	1,000	0.4

Note:
[1] Measured at 298K and 101.325kPa

3.3 Description of Environment

3.3.1 The Project encompasses the Loop and Ma Tso Lung (MTL) areas, covers mainly the rural hinterland, including rural settlements and fishponds. On the western section of the Project, it connects with the north-eastern side of the Loop as outlined in the approved Lok Ma Chau Loop Outline Zoning Plan (LMCL OZP) (OZP No. S/LMCL/2) and follows the Shenzhen Meander, traversing hilly terrain that includes Horn Hill (Ngau Kok Shan). According to the approved LMCL OZP, it aims to establish a Hong Kong-Shenzhen Innovation and Technology Park, which will strengthen the technological development in Hong Kong, focusing on scientific research, cultural and creative industries and other complementary facilities. The eastern side of the Project covers the MTL area, adjacent to Shun Yee San Tsuen, Ma Tso Lung San Tsuen and will be developed under Agreement No. CE 46/2022- Ma Tso Lung Area and Other Sites in Kwu Tung North New Development Area and North District – Feasibility Study (hereinafter referred to as “MTL Project”).

3.3.2 The nearest EPD air quality monitoring station is North monitoring station and the recent five years (2020 – 2024) averaged concentrations of air pollutants relevant to the Project are recorded and summarised in **Table 3.3**. No exceedances were recorded at the North monitoring station during the period of 2020 – 2024, except the annual FSP in 2024.

Table 3.3 Average Concentrations of Pollutants in the Recent Five Years (Year 2020 – 2024) at North EPD Air Quality Monitoring Station

Pollutant	Averaging Time		AQOs (µg/m^3)	2020	2021	2022	2023	2024
RSP	24-hr	10th Highest	75	55	62	50	57	56
	Annual		30	-	25	23	27	25
FSP	24-hr	19th Highest	37.5	29	29	28	28	29
	Annual		15	-	15	14	15	**16**
NO_2	1-hr	19th Highest	200	112	135	115	116	114
	24-hr	10th Highest	120	54	70	58	58	56

Pollutant	Averaging Time	AQOs (μg/m^3)	2020	2021	2022	2023	2024
	Annual	40	-	36	31	30	29

Notes:
[1] All data is calculated from the hourly data provided in EPD's website (http://epic.epd.gov.hk/EPICDI/air/station/?lang=en) and EPD's Smart Air Quality Platform (SAMP) v2.1 (https://aqia.epd.gov.hk/).
[2] Reference conditions of gaseous pollutants concentration data: 293K and 101.325 kPa.
[3] Since the North general station was commissioned on 10 July 2020, annual data from the North general station was available starting from 2021.
[4] Underlined and **bolded** value indicates exceedance in relevant criterion.

3.3.3 Apart from the air quality monitoring data, EPD has released a set of background levels from "Pollutants in the Atmosphere and their Transport over Hong Kong" (PATH) model (version 3.0). The PATH model is a regional air quality model developed by EPD to simulate future background air quality concentrations in Hong Kong. The Project covers 8 grids in the PATH v3.0 model which are shown in **60722948/A10/FIGURE 3.1**. As the Project is targeted for completion in Year 2031, the air quality concentrations predicted by PATH v3.0 for Year 2030 have been extracted and presented in **Table 3.4**. The predicted background concentrations of NO_2, RSP and FSP in these areas are well below the AQOs.

Table 3.4 Predicted Background Air Pollutants Concentration in Year 2030 (Extracted from the PATH v3.0)

Pollutants		RSP		FSP		NO_2		
Averaging Time		24-hr 10th Highest	Annual	24-hr 19th Highest	Annual	1-hr 19th Highest	24-hr 10th Highest	Annual
AQOs (μg/m^3)		**75**	**30**	**37.5**	**15**	**200**	**120**	**40**
PATH Grid Cell	(30,55)	55	21	31	13	88	29	15
	(30,56)	56	22	33	13	100	38	20
	(31,55)	54	20	31	13	79	25	13
	(31,56)	54	21	32	13	90	30	15
	(32,55)	56	21	33	13	76	23	13
	(32,56)	54	21	32	13	84	27	14
	(33,55)	54	21	31	13	71	22	12
	(33,56)	55	21	31	13	81	27	13

3.4 Identification of Air Sensitive Receivers

3.4.1 In accordance with Annex 12 of the EIAO-TM, any domestic premises, hotel, hostel, hospital, clinic, nursery, temporary housing accommodation, school, educational institution, office, factory, shop, shopping centre, place of public worship, library, court of law, sports stadium or performing arts centre are considered as air sensitive receivers (ASRs). Any other premises or place with which, in terms of duration or number of people affected, has a similar sensitivity to the air pollutants as the aforelisted premises and places is also considered to be a sensitive receiver.

3.4.2 In accordance with the EIA Study Brief, the assessment area for air quality assessment should be defined by a distance of 500 m from the boundary of the Project and its associated works. Illustration of the proposed assessment area is presented in **60722948/A10/FIGURE 3.1** and **3.1a to 3.1e**.

3.4.3 For identification of the representative ASRs within the assessment area that would likely be affected by the potential impacts from the construction and operation of the Project, a review has been conducted based on relevant available information including topographic maps, Outline Zoning Plans (OZPs) and other published plans in the vicinity of the Project site. The representative ASRs within the assessment area are presented in **Table 3.5** and illustrated in **60722948/A10/FIGURE 3.1**, **3.1a** to **3.1e**. It is confirmed that the selected ASRs listed in **Table 3.5** are representative due to their proximity to the Project for this assessment, based on the best available information at the time of the EIA report.

Table 3.5 Representative Air Sensitive Receivers for Air Quality Impact Assessment

ASR ID	Description	Land Use [1]	Approx. Distance between ASR and the Project Boundary (m)	Assessment Height Above Ground (mAG) [2]
Existing ASR				
A1	Temporary Structure to the South of Lamp Post VG3160	R	5	1.5 – 5
A2	Lok Ma Chau Operation Base Lok Ma Chau Division Hong Kong Police Force	O	5	1.5 – 5
A3	Village House in DD96 1427	R	65	1.5 – 10
A4	Village House to the Southwest of Lamp Post VG4766	R	75	1.5 – 10
A5	Village House to the West of Lamp Post VG4766	R	60	1.5 – 10
A6	Village House to the North of Lamp Post VG3124	R	115	1.5 – 5
A7	Temporary Structure to the Northeast of Lamp Post VA4082	R	20	1.5 – 5
A8	Village House to the Southeast of Lamp Post EC1009	R	75	1.5 – 10
A9	Village House in DD96 1283	R	90	1.5 – 5
A10	11 Ma Tso Lung Shun Yee San Tsuen Block C	R	55	1.5 – 10
A11	Village House to the South of Lamp Post N4004	R	35	1.5 – 10

ASR ID	Description	Land Use [1]	Approx. Distance between ASR and the Project Boundary (m)	Assessment Height Above Ground (mAG) [2]
A12	1 Ma Tso Lung Shun Yee San Tsuen Block E	R	5	1.5 – 10
A13	Temporary Structure to the East of Lamp Post N4001	R	10	1.5 – 5
A14	Temporary Structure to the South of Lamp Post CD0749	R	10	1.5 – 5
A15	1 Ma Tso Lung Shun Yee San Tsuen Block A	R	20	1.5 – 5
A16	6 Ma Tso Lung Shun Yee San Tsuen Block A	R	20	1.5 – 5
A17	2 Ma Tso Lung Shun Yee San Tsuen Block B	R	25	1.5 – 5
A18	1 Ma Tso Lung Shun Yee San Tsuen Block C	R	25	1.5 – 5
A19	4A Ma Tso Lung Shun Yee San Tsuen Block C	R	20	1.5 – 10
A20	1 Ma Tso Lung Shun Yee San Tsuen Block D	R	15	1.5 – 5
A21	5 Ma Tso Lung Shun Yee San Tsuen Block D	R	10	1.5 – 5
Planned ASR				
PA01 [3]	Planned Area for Research & Development, Higher Education and Cultural & Creative Uses at the Loop	OU	15	1.5 – 30
PA02 [3]	Planned Area for Research & Development, Higher Education and Cultural & Creative Uses at the Loop	OU	15	1.5 – 50
PA03 [3]	Planned Fire Station & Ambulance Depot, HKPF Facilities and Possible Boundary Crossing Facilities at the Loop	G/IC	135	1.5 – 30
PA04 [3]	Planned Commercial Uses at the Loop	C	180	1.5 – 50

ASR ID	Description	Land Use [1]	Approx. Distance between ASR and the Project Boundary (m)	Assessment Height Above Ground (mAG) [2]
PA05 [3]	Planned Area for Research & Development, Higher Education and Cultural & Creative Uses at the Loop	OU	240	1.5 – 50
PA06 [3]	Planned Commercial Uses at the Loop	C	310	1.5 – 50
PA07 [3]	Planned Fire Station & Ambulance Depot, HKPF Facilities and Possible Boundary Crossing Facilities at the Loop	G/IC	395	1.5 – 30
PA08 [3]	Planned Residential Development at Ma Tso Lung Development Res6	R	Within Project Boundary	1.5 – 140
PA09 [3]	Planned Open Space at Ma Tso Lung Development O4	OS	15	1.5
PA10 [3]	Planned Open Space at Ma Tso Lung Development O4	OS	Within Project Boundary	1.5
PA11 [3]	Planned Open Space at Ma Tso Lung Development O4	OS	35	1.5
PA12 [3]	Planned Residential Development at Ma Tso Lung Development Res3	R	20	1.5 – 140
PA13 [3]	Planned Residential Development at Ma Tso Lung Development Res4	R	95	1.5 – 145
PA14 [3]	Planned Residential Development at Ma Tso Lung Development Res2	R	160	1.5 – 150
PA15 [3]	Planned Residential Development at Ma Tso Lung Development Res1	R	395	1.5 – 145
PA16 [3]	Planned Residential Development at Ma Tso Lung Development Res5	R	295	1.5 – 150

Notes:
[1] R – Residential, O – Office, OS – Open Space, OU – Other Specified Uses, C – Commercial, G/IC – Government, Institution or Community
[2] The lower bound of assessment shown in the table in the lowest level with air-sensitive uses, full range of assessment heights have been covered for each ASR. For all planned ASRs, the lowest assessment height is assumed to be 1.5mAG.
[3] Since the population intake for PA01-PA07 is scheduled for after the year 2031, and for PA08-PA16 in the year 2034, the ASRs (PA01 – PA16) will not be considered as ASRs during the construction phase from 2027 to 2031.

3.5 Identification of Environmental Impact

Construction Phase

3.5.1 According to the construction programme, the construction works of the Project is tentative scheduled to commence in Q1 / Q2 2027 and completed by Q4 2031. Air quality impact may arise from various construction activities of the Project, including:

- site clearance and site formation works;
- the use of construction plant equipment (i.e. Non-road Mobile Machinery (NRMM)) and construction trucks;
- slopeworks;
- construction of shallow underpass section connecting the Loop internal road to Eastern Connection Road across the meander and Ecological Area;
- construction of viaduct and at-grade road sections; and
- wind erosion of exposed work area.

3.5.2 Most of the construction activities would involve excavation, spoil handling and backfilling of different scales. These dusty construction activities and wind erosion of exposed sandfill areas would cause potential fugitive emission in particulates.

3.5.3 Also, the Project comprises of underpass/depressed road section across the meander, which would be constructed with cut-and-cover method. Potential sediment of estuarine deposits, marine deposits and pond sediment would need to be excavated. The excavation and handling of sediment may cause potential odour impacts during construction.

3.5.4 Vehicular emissions from induced construction traffic for the transportation of spoils/excavated materials would cause potential gaseous and particulates emission. On-site use of diesel-powered engines is also the potential source of particulate matters (PM) and gaseous pollutants, such as NOx, SO_2, CO and smoke.

Concurrent Projects

3.5.5 As stated in **Section 2.10**, there are concurrent projects identified within 500m of the Project. The construction works of the Project will be tentatively commenced in Q1 / Q2 2027 and be completed by Q4 2031. All planned and committed concurrent projects are showed in **60722948/A10/FIGURE 2.9** and summarised in **Table 3.6** below:

Table 3.6 Summary of Concurrent Projects

Project	Tentative Construction Programme	Approximate Separation Distance from the nearest Project Site Boundary (m)
Development of the Loop – Main Works Package 1	2021 – 2027	Within the Project boundary

Project	Tentative Construction Programme	Approximate Separation Distance from the nearest Project Site Boundary (m)
Development of Hong Kong – Shenzhen Innovation and Technology Park at the Loop	To be completed by 2035 [1]	Within the Project boundary
MTL Project	2028 – 2034	Within the Project boundary
Northern Metropolis Highway – Kwu Tung Section	Commission in about 3 to 4 years after the commissioning of the San Tin Section, in or before 2036.	40

Note:
[1] The timeline mentioned refers to Chapter 8 of the "*Development Outline for the Hong Kong Park of the Hetao Shenzhen-Hong Kong Science and Technology Innovation Co-operation Zone*," For further details, please refer to the document - https://www.itib.gov.hk/assets/files/%E6%B2%B3%E5%A5%97%E6%B7%B1%E6%B8%AF%E7%A7%91%E6%8A%80%E5%89%B5%E6%96%B0%E5%90%88%E4%BD%9C%E5%8D%80%E9%A6%99%E6%B8%AF%E5%9C%92%E5%8D%80%E7%99%BC%E5%B1%95%E7%B6%B1%E8%A6%81%20(EN).pdf).

3.5.6 Development of the Loop – Main Works Package 1 (hereinafter referred to as "LMCL project") comprises the development and infrastructure within the Loop and the adjacent areas in Hong Kong. The construction of LMCL project is commenced in 2021 for expected completion in phases in 2027. The main works package 1 includes the site clearance and formation within the Loop, and engineering infrastructure works supporting Phase 1 development of the Hong Kong Shenzhen Innovation and Technology Park (hereinafter referred to as "the Park"), including Western Connection Road and Direct Road Link for connection with the Loop. According to the available information, the Park is expected to be fully completed by 2035, cumulative construction air quality impacts may arise.

3.5.7 MTL project is developed the sites for residential and employment uses, supporting G/IC facilities, open space and/or amenities. Based on the best available information at the time of this EIA study, the construction works of the project is tentatively scheduled to commence in 2028 and the first phase of population intake will be scheduled in 2034. As the planned construction period of MTL project overlapped with the Project, cumulative construction air quality impacts may arise.

3.5.8 Northern Metropolis Highway (hereinafter referred to as "NMH") is a major east-west corridor that connects various developments, including Ngau Tam Mei, San Tin Technopole, Kwu Tung North/Fanling North and New Territories North (NTN) New Town). NMH consists of four road sections – Tin Shui Wai section, San Tin section, Kwu Tung section and NTN New Town section. Kwu Tung Section is situated on the southeastern side of the Project. According to the best available information at the time of this EIA study, construction of the NMH San Tin Section is expected to commence by 2036, with subsequent phases of the remaining sections planned for completion within 3 to 4 years after the San Tin Section is commissioned. The planned construction period of NMH (Kwu Tung Section) will not overlap with the Project, subject to the construction programme of NMH. Cumulative construction air quality impacts associated with NMH Kwu Tung Section are not anticipated.

Operation Phase

Vehicular Emission from Proposed and Other Roads

3.5.9 Vehicular emission from the proposed and other roads including Ma Tso Lung Road, Liu Pok Road and Border Road within 500m assessment area contributes to the ambient air quality. These emissions include open road emissions, portal and top opening emissions associated with the underpass and depressed sections. The locations of these sources are shown in **60722948/A10/FIGURE 3.3**.

Vehicular Emission associated with Public Transport Interchange (PTI) and Minibus Terminus

3.5.10 Facilities with frequent operation associated with vehicles also contribute to ambient air quality by vehicular running, idling and start emission within these facilities of concern. According to the desktop study and site survey conducted on 16 April 2024, 14 August 2024 and 14 January 2025, one mini-bus terminus in Shun Yee San Tsuen is located within 500m assessment area. Based on the available information at the time of EIA preparation, two planned PTIs are located in the proposed Development of MTL Area and the Loop. According to the study conducted by Hong Kong-Shenzhen Innovation and Technology Park Limited (HSITPL), it is assumed that all road-based public transport serving the Loop will adopt Electric Vehicles (EVs) or clean-energy vehicles, with zero air pollutant emissions. In view of this assumption, formal permission was requested from HSITPL to use the relevant data, and they raised no objections to the adoption of electric vehicles (EVs) or clean energy within the Loop for the purpose of this EIA study. As such, no starting, idling and running emissions associated with the PTI at the Loop. Therefore, the planned PTI located in the proposed Development of MTL Area and the one existing minibus terminus on Ma Tso Lung Road are considered in the assessment, and their locations are illustrated in **60722948/A10/FIGURE 3.3**.

Major Point Sources within 4km

3.5.11 Chimney survey conducted on 16 April 2024, 14 August 2024, and 14 January 2025. In addition, a review of the information from Hong Kong Environmental Database (HKED) published by EPD and the approved EIA report (AEIAR-176/2013) was also undertaken. Based on the best available information, no existing, planned or committed industrial emission sources or chimneys were identified within 500m assessment area, including the portion of the Loop within the assessment area. However, two major point sources located within 4km of the site boundary are identified on CED as shown in **60722948/A10/FIGURE 3.4**, namely Asphalt Plant from K. Wah Asphalt Limited (AP-1) and Organic Resources Recovery Centre Phase 2 (OPARK2).

Identification of Key Air Pollutants during Operation Phase

3.5.12 Vehicular emission will be the dominant source of air pollutants arising from the operation of this Project. The key air pollutants associated with vehicular emission during operation phase include NOx, RSP and FSP.

3.6 Assessment Methodology

Construction Phase

3.6.1 Considering the small scale and nature of the construction activities for the Project, which would be carried out in phasing or sections, the potential air quality impact and odour impact during the construction phase should be limited. With the implementation

of mitigation measures stipulated in Air Pollution Control (Construction Dust) Regulation, careful scheduling of works, and close liaison with contractors of concurrent works to avoid overlapping of heavy or dusty construction works as far as practicable, and the implementation of good site practices recommended in **Sections 3.8.1** to **3.8.4**, construction air quality impact and odour impact will be controlled. Therefore, a qualitative assessment approach is adopted for construction air quality impact assessment in accordance with Annex 12 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) to ensure that the Air Pollution Control (Construction Dust) Regulation is complied with.

Operation Phase

Vehicular Emissions from Open Roads

3.6.2 Open section of existing roads (e.g. Ma Tso Lung Road, Liu Pok Road and Border Road), proposed and planned road networks (including at-grade roads, depressed roads, and viaduct) within the assessment area were included in the assessment using EPD Smart Air Modelling Platform v2.1 (SAMP v2.1) according to the latest Technical Notes on Air Quality Modelling on EPD's website and the EPD's User Guide for the ROAD Tool in Smart Air Modelling Platform. The traffic flow induced from the LMCL project, MTL project, the Park and NMH Kwu Tung Section is considered in the assessment. The modelled fleet is broken down into 18 vehicle classes based on the information in the Appendix 1 of Guideline on Modelling Vehicle Emissions published by EPD. The predicted 24-hour traffic flow and vehicle compositions at the identified roads was provided by the traffic consultant according to the methodology endorsed by Transport Department (TD) and are presented in **Appendix 3.1**.

3.6.3 EMFAC-HK v4.3 model was adopted in SAMP v2.1 to estimate the vehicular emission rates of NOx (i.e. initial NO + initial NO_2), RSP and FSP for vehicular emission arising from open road, and the adopted temperature and relative humidity data were extracted from the meteorological output in SAMP v2.1 of the PATH grids where the road segments are located. The monthly hour minimum temperature and relative humidity data was adopted for the prediction of short-term averages of NO_2, while the monthly hour average temperature and relative humidity data was adopted for the prediction of long-term averages of NO_2, and annual minimum temperature and relative humidity was adopted for prediction of short-term and long-term averages of RSP and FSP.

3.6.4 Based on on-site observations and meter parking data from the TD, there are no heavy goods vehicles or non-franchised buses parked on the streets within the assessment area. To prevent any underestimation, a conservative 'broad-brush' approach is adopted for estimating start emissions from all vehicle types allocating emissions to local/rural roads. For the broad-brush approach, start emissions for each vehicle class are assumed directly proportional to Trips per Vehicle-kilometre-travelled ratio (Trip/VKT) with consideration of the proportion of local and rural roads within Hong Kong based on the Annual Traffic Census (ATC) prepared by the Transport Department (TD). For SAMP v2.1, default trip and VKT values from EMFAC-HK model v4.3 are used in the calculations and the proportion of local and rural roads within Hong Kong of 21.8% is calculated based on raw data from ATC 2023 provided by TD in accordance with the methodology described in the Appendix K of ATC 2023, with details provided in **Appendix 3.2**. All rural and local roads within the assessment area are considered using the broad-brushed approach, including roads leading to the identified HGV parking sites. Zero Emission Vehicle (ZEV) Scenario has been considered in the open road emission calculation. The relevant parameters utilised in this assessment are provided in **Appendix 3.3.**

3.6.5 With reference to **Chapter 4**, the findings of traffic noise impact assessment indicates that noise barriers (e.g., vertical or cantilever barriers) will not be considered in the Project. Additionally, there are no existing noise barriers within 500m assessment area. Therefore, the air quality impact associated with the noise vertical barriers and cantilever barriers is not anticipated.

Portal Emission and the Top Opening from the underpass of the Proposed Road

3.6.6 The portal emissions (initial NO, initial NO_2, RSP, and FSP) from the shallow underpass of the Project, as well as emissions from the top openings of the shallow underpass, are considered during the operation phase.

3.6.7 For calculations of portal emissions and emissions from the top opening of the shallow underpass, temperature and RH data were extracted from the SAMP of the concerned PATH grids. The annual minimum temperature and RH values (7°C and 14%) across all concerned PATH grids were used in the calculation as a conservative approach. The SAMP v2.1 with ZEV scenario was used for the calculation of composite vehicular emission factor and emission rates. The location of portal emissions is presented in **60722948/A10/FIGURE 3.3** and the detailed calculation are presented in **Appendix 3.4**. According to the Permanent International Association of Road Congress Report (PIARC, 2019), pollutants were assumed to eject from the portal as a portal jet such that 2/3 of the total emissions were dispersed within the first 50 m of the portal and 1/3 of the total emissions within the second 50 m.

3.6.8 The shallow underpass is equipped with two top openings, each measuring ~1,100 m² in area and approximately 50 m in length. For a conservative estimate, it is assumed that the emissions from each top opening are equal to the portal emissions along the entire shallow underpass. As a result, the total emissions from the shallow underpass amount to 3 times the portal emissions, accounting for emissions distributed from both top two openings as well as the portal itself.

Emissions associated with the existing mini-bus terminus at Shun Yee San Tsuen and the Planned PTIs in the proposed MTL Project and within the Loop

3.6.9 The start emissions, running exhaust emissions, and idling emissions (i.e., NO, NO_2, RSP, and FSP) associated with the mini-bus terminus in Shun Yee San Tsuen (E1) and the planned PTI at MTL (E2) are calculated based on the start emission and running exhaust emission factors predicted by EMFAC-HK model, cold idling emission factors from *Calculation of Start Emissions in Air Quality Impact Assessment* published by EPD, warm idling emission factors from *Road Tunnels: Vehicle Emissions and Air Demand for Ventilation published by World Road Association.* According to the findings of 24-hour site survey carried out on a normal working day, there is no terminating mini-bus and no overnight parking in Shun Yee San Tsuen (E1). For E2, since site survey cannot be conducted on the planned PTI, reference has been made to the existing PTI of similar nature that primarily serve residential sites in the North district. Specifically, Queen's Hill PTI (with bus route 78A) and Luen Wo Hui PTI (with bus routes 277E and 277X) were referenced. Soaking times at these two locations generally range from 10 to 20 minutes. According to the latest draft Transport and Traffic Impact Assessment (TTIA) report of MTL Development, the planned PTI will accommodate 6 bus routes with approximately 60 bus departures during the peak period.

3.6.10 Start emissions for diesel vehicles fitted with selective catalytic reduction (SCR) devices vehicles were adjusted based on the idling emission and would be released over total spread distance of 700m from where the start takes place, while running exhaust and

idling emissions would be released on the spot. Running exhaust and idling emissions from terminating and non-terminating vehicles, and adjusted start emission from terminating vehicles are considered for emissions inside terminus / PTI while the remaining adjusted start emission from terminating vehicles is considered for emissions outside terminus / PTI. The temperature and RH values for all concerned PATH grids were compared and the minimum temperature and RH values (7ºC and 14%) were adopted in the calculation as a conservative approach. The location of emission sources and the detailed calculation of the emission are presented in **Appendix 3.5** and **60722948/A10/FIGURE 3.3**.

Determination of Assessment Year for Vehicular Emission

3.6.11 The vehicular emission burdens of NOx, RSP and FSP for all existing and proposed roads within 500m assessment area from commissioning year to 15 years after commissioning, namely 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041 and 2046, are generated from SAMP v2.1 with ZEV scenario adopted and are shown in **Appendix 3.2**. The worst assessment year is determined by the highest vehicular emission burden among NOx, RSP and FSP. According to the result in **Appendix 3.2**, the assessment year for NOx, RSP and FSP is Year 2034.

Existing Industrial Emission and Major Point Sources within 4km

3.6.12 Given the complex terrain and hilly landscape between the Project site and the two major point sources (AP-1 and OPARK2), emissions from these sources would be obstructed by Crest Hill. Consequently, there would be no direct impact from AP-1 and OPARK2 on the identified ASRs. Therefore, AP-1 and OPARK2 were not included in the air quality modelling as background emissions.

Dispersion Modelling and Modelling Approach

3.6.13 With reference to Section 3.5.3, Appendix B of the EIA Study Brief, and the EPD's *Guidelines for Local-Scale Air Quality Assessment Using Models*, the AERMOD model was employed to predict the potential air quality impact.

3.6.14 All open road vehicular emissions were modelled as "LINE" sources. The hourly initial NO, NO_2, RSP and FSP emission factors of each road segment in grams per second per square meter were generated using SAMP v2.1. **Appendix 3.3** shows the model parameters in detail.

3.6.15 Since the mini-bus terminus in Shun Yee San Tsuen (E1) is in open-area and does not have any mechanical ventilation for exhaust, the emissions within parking site are modelled as "AREAPOLY" sources according to the minibus stop observed on-site. There is no terminating minibus in E1, thus no starting emission and spread distance for E1. For E2, considering the study of MTL Development is still on-going, the detailed design of the planned PTI is not available at the time of preparation of this study. Since E2 is located under the residential building, it is assumed that enclosed design with mechanical ventilation for exhaust is adopted, all emissions generated within the planned PTI will be allocated in the egress of the PTI and modelled as "VOLUME" source at that location as a conservative approach. **Appendix 3.5** shows the model parameters in detail.

3.6.16 All portal emissions were modelled as VOLUME source and emission from the top opening along the proposed ECR in shallow underpass section were modelled as AREA source.

3.6.17 Hourly meteorological conditions including wind data, temperature, relative humidity, pressure, cloud cover and mixing height are extracted from the WRF meteorological data adopted in the PATH v3.0 model, obtained through SAMP v2.1. The meteorological data was inputted as on-site data into AERMET (the meteorological pre-processor of AERMOD). Surface characteristic parameters such as albedo, Bowen ratio and surface roughness are required in the AERMET (the meteorological pre-processor of AERMOD) and could be found in **Appendix 3.6**. The land use characteristics of the project area are customised according to the latest design of the related projects, while that of the surrounding are classified based on the Planning Department's 2023 land utilisation map, and the AERMET output are generated by SAMP v2.1 for the AERMOD model run. Elevated terrain and urban mode in AERMOD were adopted for this assessment, with an urban population of 342,600 according to the 2024 Mid-year Population by District Council district released by Census and Statistics Department.

Cumulative Impact

3.6.18 The PATH model output was added to the sum of the AERMOD model results sequentially on an hour-to-hour basis to derive the short-term and long-term cumulative impacts at the ASRs. Year 2030 PATH model output was adopted as the background concentrations as conservative approach and the assessment year is 2034.

3.6.19 Time-averaged results, namely hourly, daily and annual, will be derived from the cumulative hour-by-hour results. For annual average, the sum of all hourly concentrations is divided by the number of hours during the year to obtain the annual-averaged concentration. The air quality impact upon ASRs will be evaluated by respective concentration limits specified in the AQO. For daily average, cumulative daily-averaged results at each ASR amongst 365 days are ranked by the concentration and the ranked daily-averaged concentrations are compared with the maximum allowable concentration to determine the number of exceedances throughout a year. The air quality impact upon ASRs was evaluated by number of exceedances per annum against the number of exceedances allowed as specified in the AQO or EIAO-TM for hourly and daily averaged result.

3.6.20 As suggested by "Guidelines on Assessing the 'TOTAL' Air Quality Impacts", an integrated modelling system, PATH v3.0 which is developed and maintained by EPD is applied to estimate the future background pollutant concentrations. The emission sources including those in Pearl River Delta Economic Zone, roads, marine, airport, power plants and industries within Hong Kong are all considered in the PATH v3.0 model. The year of the adopted PATH background data is 2030, which has the highest background concentration within the assessment periods. The assessment area covers 8 grid cells of PATH v3.0, namely grid (30,55), (30,56), (31,55), (31,56), (32,55), (32,56), (33,55) and (33,56).

Conversion of NO_x to NO_2

3.6.21 The conversion from NO_x to NO_2 is necessary to compare the operation phase cumulative results against AQOs. The conversion method follows the approaches stated in the 'Guidelines on Choice of Models and Model Parameters' published by EPD. Two approaches, namely Ozone Limiting Method and Jenkin Method, were adopted for the conversion from NO_x to NO_2.

Ozone Limiting Method for Short-term Cumulative NO_2 Assessment

3.6.22 For the short-term cumulative NO_2 assessment (i.e., predictions of hourly and daily average NO_2 concentration), The predicted initial NO concentrations from vehicle-related sources (i.e., emissions from open road, portals, top openings, minibus terminus at Shun Yee San Tsuen, the planned PTI at the proposed MTL Development) and 90% of the predicted NOx concentrations from industrial sources will firstly be added together on an hour-to-hour basis and Ozone Limited Method (OLM) will be applied subsequently to calculate NO_2 based on the predicted O_3 level from PATH v3.0. According to the Heathrow Airport EIA report, the initial NO_2/NOx ratios of marine and industrial emission sources are 10%. The NO_2/NOx conversion was calculated as follows:-

$[NO_2]_{predicted} = [NO_2]_{vehicular} + 0.1 \times [NO_x]_{industrial} + MIN \{[NO]_{vehicular} + 0.9 \times [NOx]_{industrial}, or (46/48) \times [O_3]_{PATH}\}$

where

$[NO_2]_{predicted}$	is the predicted NO_2 concentration
$[NO_2]_{vehicular}$	is the sum of predicted initial NO_2 concentration from open roads, portals, minibus terminus and planned PTI
$[NO_x]_{industrial}$	is the sum of predicted initial NO_x concentration from the industrial emission sources
$[NO]_{vehicular}$	is the sum of predicted initial NO concentration from open roads, portals, minibus terminus and planned PTI
MIN	means the minimum of the two values within the brackets
$[O_3]_{PATH}$	is the representative O_3 from rerun PATH concentration (from other contribution)
(46/48)	is the molecular weight of NO_2 divided by the molecular weight of O_3

Jenkin Method for Long-term Cumulative NO_2 Assessment

3.6.23 For the long-term cumulative NO_2 assessment (i.e., predictions of annual average NO_2 concentration). Jenkin method was adopted for the conversion of cumulative NO_x to NO_2 by using the functional form of an annual mean of NO_2-to-NO_x with reference to the *Review of Methods for NO to NO_2 Conversion in plumes at short ranges*[1]. The mentioned functional form is referenced from Jenkin, 2004a[2] and is presented as follows:

$$[NO_2] = \frac{\left([\mathrm{NOx}] + [\mathrm{OX}] + \frac{\mathrm{J}}{\mathrm{k}}\right) - \sqrt{([NOx] + [OX] + \frac{J}{k})^2 - 4[NOx][OX]}}{2}$$

where

$[NO_2]$	is the NO_2 concentration
$[NO_x]$	is the NO_x concentration
[OX]	is the sum of NO_2 concentration and O_3 concentration (i.e. $[OX] = [NO_2] + [O_3]$)

[1] *Environment Agency. 2007.Review of methods for NO to NO_2 conversion in plumes at short range. Prepared by Environmental Agency.*
[2] *Jenkin. 2004a. Analysis of sources and partitioning of oxidant in the UK – Part 1: The NOx-dependence of annual mean concentrations of nitrogen dioxide and ozone.* Atmospheric Environment, 38, 5117-5129.

J is the photolysis rate of NO_2

k is the rate coefficient for reaction between NO and O_3

3.6.24 The annual mean data obtained from the EPD's air quality monitoring stations (AQMS), including the North general station, Tap Mun general station, and three roadside stations (Causeway Bay, Central, and Mong Kok), were analysed using the functional form provided above. The North general station was chosen as the representative station due to its proximity to the Project. However, since the North general station was commissioned on 10 July 2020, data from this station was also available starting from 2021. The Tap Mun general station and three roadside stations were included to cover a wider range of NOx concentration. The function from curve would fit the annual mean data when [OX] = 95.57 μg/m^3 and J/K = 17.114 μg/m^3. The obtained functional form curve was adopted for the cumulative annual average NO_x to NO_2 conversion. As shown in **Appendix 3.7**, the curve is slightly higher than all the annual mean data obtained from AQMS and on-site measurement data, underestimation of the annual average NO_2 concentration is not expected. The cumulative annual average NO_x to NO_2 conversion equation for this assessment was calculated as follows:-

$$[NO_2]_c = \frac{([NOx]_c + 95.57 + 17.114) - \sqrt{([NOx]_c + 95.57 + 17.114)^2 - 4[NOx]_c \times 95.57}}{2}$$

where

$[NO_2]_c$ is the predicted cumulative NO_2 concentration in μg/m^3

$[NO_x]_c$ is the predicted cumulative NO_x concentration (i.e., the sum of the total predicted NO_x concentration from AERMOD and PATH v3.0) in μg/m^3

3.7 Prediction and Evaluation of Environmental Impacts

Construction Phase

3.7.1 With reference to **Section 2.9** and **60722948/A10/FIGURE 2.8**, the Project will be implemented in three main sections: Section 1 which extends from the Loop to the end of the underpass near the Lok Ma Chau Operation Base; Section 2 which includes from the depressed road near the Operation Base to the viaduct section near Shun Yee San Tsuen; and Section 3, which covers from the at-grade alignment near Shun Yee San Tsuen to the viaduct near Ma Tso Lung San Tsuen. Construction will be carried out in two phases, with Phase 1 comprising Sections 1 and 3, and Phase 2 comprising Sections 2 and 3. Section 3 will be undertaken throughout the entire construction period, while Sections 1 and 2 will not be undertaken concurrently. The shallow underpass and depressed roads across the existing meander and ponds in Section 1 will adopt the cut-and-cover method, which is anticipated to generate air emissions and odour impact. Referring to **Chapter 6**, the total volume of sediment generated is estimated to be approximately 100,700 m^3, and the construction works would be conducted separately in different works zone within Section 1 as shown in **60722948/A10/FIGURE 3.2b**. For the viaduct structure, which extends from the Ma Lung Tso Blue House, crosses Ma Tso Lung Road, and terminates at the hillside in northern Shun Yee San Tsuen, dusty activities will include minor excavation and limited backfilling for each column installation. These activities will occur within a limited area (as illustrated in **60722948/A10/FIGURE 3.2a**) of approximately 400 m². The shortest distance between the dusty activities (i.e. column installation) and the ASR (i.e. A21) is approximately 14 m. While the construction of at-grade road sections would be conducted at the same phase, except three works zones showed in **60722948/A10/FIGURE 3.2b**. The

construction works of each zone would be conducted separately, and the estimated works area would be around 6,000 m^2 for each. To minimise the impact of construction activities on nearby ASRs, construction will not occur simultaneously across the entire work area. Instead, it will be conducted in phases or sections during different periods. Air emissions from these activities are expected to be controlled and localised. The precast concrete method will be employed for the viaduct, eliminating the need for an on-site concrete batching plant, thus limiting air emissions. For the at-grade road, an open-cut excavation method will be used. The amount of inert materials generated during the construction phase is expected to be around 407,900 m^3 and approximately 116,400 m^3 is expected to be reused on-site as backfill material. With the implementation of good site practices and appropriate mitigation measures outlined in **Sections 3.8.1** to **3.8.4**, the construction air quality impact and odour impact could be controlled.

3.7.2 As mentioned in **Sections 3.5.1** to **3.5.4,** the small scale and nature of the construction activities for the Project, which would be carried out in phasing or sections, the potential air quality impact during the construction phase would be limited. Air quality control measures stipulated in Air Pollution Control (Construction Dust) Regulation should be implemented as far as practicable to abate the air emission from the construction sites. Regular watering should be provided at the excavation and backfilling works, spoil handling and exposed areas. Stockpile areas should be covered with impervious sheets as far as practicable. Haul roads should be paved and regularly wetted to suppress the fugitive dust emission caused by the travelling construction vehicles. Vehicles transporting dusty spoil should be properly covered with mechanical cover or tarpaulin sheets to avoid any dust pick up by gust during travel. Wheel washing facility should also be provided at each exit of construction site such that no residue on the body of the construction vehicles would cause dust emission on public roads. In addition, slope works will be conducted along the hillside in the northern Shun Yee San Tsuen and Ping Hang, near Border Road. The areas requiring slopeworks are limited to around 12,300 m^2. Therefore, potential fugitive dust emission associated with the slope works are expected to be controlled and localised due to limited space for construction. Good site practices recommended in **Sections 3.8.1** to **3.8.4** would be implemented. With the implementation of appropriate air quality control measures and good site practices, the air emission from the construction works would be controlled.

3.7.3 According to **Section 6.6**, a maximum of 31 dump trucks single-trips per day, will be needed to transport inert C&D materials. This will result in increased vehicular emissions and particulate matter along the transport route. Therefore, the transportation route should avoid local roads, and truck traffic should be scheduled outside of peak hours. Additionally, each dump truck will be equipped with a watertight container and a mechanical cover to prevent dust and material leakage. With the implementation of these mitigation measures, it is anticipated that no adverse air quality impact would be caused by the transportation of C&D materials.

3.7.4 Fuel combustion from the use of powered mechanical equipment (PME) and construction / dump trucks during construction works could be a source of PM, NO_2, SO_2 and CO. It is anticipated that with a maximum of 19 number of PMEs and a maximum of 31 dump truck trips per day would be in operation during construction phase. With the implementation of the requirements and measures stipulated in the *Air Pollution Control (Non-Road Mobile Machinery (NRMM)) (Emission) Regulation* and *Air Pollution Control (Fuel Restriction) Regulations,* the emissions from construction equipment and vehicles are not expected to cause adverse air quality impact to the surrounding ASRs. Fuel with sulphur content not exceeding 0.001% by weight will be

used to minimise SO_2 emission in accordance with the *Air Pollution Control (Fuel Restriction) Regulations*.

3.7.5 Air quality control measures are recommended in **Section 3.8.1** to **3.8.4**, and a comprehensive EM&A programme would be implemented to ensure that the nearby ASRs would not be subject to adverse air quality impact and odour impact during the construction stage.

Concurrent Projects

3.7.6 The construction of the Development of the Loop Project commenced in 2021 and is expected to be completed by 2027, while the Park is tentatively scheduled for full completion by 2035. There may be an overlap between the construction areas of the Park and the LMCL project. To minimise the occurrence of concurrent dusty / heavy activities, close coordination will be maintained among the contractors of the development of the Loop project, the Park, and ECR. By implementing good site practices, adverse cumulative construction air quality impacts from development of the Loop Project and the Park will not be anticipated.

3.7.7 While the planned construction period of MTL project overlapped with the Project, the CEDD shall closely liaise with the corresponding parties of these two concurrent projects to avoid dusty / heavy construction works carried out concurrently. With the implementation of mitigation measures such as good site practice and close liaison with the contractors of MTL, adverse cumulative construction air quality impact arising from construction activities associated with MTL project are not anticipated.

3.7.8 The aforementioned concurrent projects have undergone EIAO processes and should have regard to the attainment and maintenance of an acceptable air quality during the construction and operation period. An EM&A programme should also be implemented during the construction stage to ensure that the air quality control measures will be implemented in order and adverse air quality impact will not be imposed. Hence, these concurrent projects will not be expected to cause adverse cumulative construction air quality impacts on the surrounding ASRs.

Operation Phase

3.7.9 The cumulative air quality impacts at the representative ASRs have been evaluated. The predicted cumulative NO_2, RSP and FSP concentrations at the representative ASRs during the operation of the Project are presented in **Table 3.7**. Detailed assessment results of all relevant assessment heights of the identified representative ASRs are provided in **Appendix 3.8**. The prediction showed that the 10th highest daily average and the annual average of RSP concentrations, the 19th highest daily average and the annual average of FSP concentrations and the 19th highest hourly average, 10th highest daily average and the annual average of NO_2 concentrations at all representative ASRs would comply with the AQOs.

Table 3.7 Predicted Cumulative Concentrations at Representative Air Sensitive Receivers During the Operation Phase

ASR	NO_2 Concentration (μg/m^3)			RSP Concentration (μg/m^3)		FSP Concentration (μg/m^3)	
	19th Highest Hourly Average	10th Highest Daily Average	Annual Average	10th Highest Daily Average	Annual Average	19th Highest Daily Average	Annual Average
AQOs	**200**	**120**	**40**	**75**	**30**	**37.5**	**15**
Existing ASRs							
A1	105 - 117	41 - 49	16 - 17	54	21	31	13
A2	92 - 95	31 - 32	15 - 16	56	21	33	13
A3	86 - 88	27	13	56	21	33	13
A4	85 - 88	26 - 27	13	56	21	33	13
A5	85 - 88	26 - 27	13	56	21	33	13
A6	90 - 91	29	14	55	21	32	13
A7	95 - 96	32	15	55	21	32	13
A8	88	27	13 - 14	56	21	33	13
A9	87 - 89	27	13	56	21	33	13
A10	89 - 90	27	14	56	21	33	13
A11	89	26 - 27	14	56	21	33	13
A12	90	26 - 28	13 - 14	56	21	33	13
A13	83	26	13	54	21	31	13
A14	82	26	13	54	21	31	13
A15	93	29 - 30	15 - 16	56	21	33	13
A16	92	29	15	56	21	33	13
A17	90	28 - 29	15	56	21	33	13
A18	90	28 - 29	15	56	21	33	13
A19	90	28 - 29	14 - 15	56	21	33	13
A20	90	28 - 29	15	56	21	33	13
A21	90	29	14	56	21	33	13
Planned ASRs							
PA01	82 - 111	26 - 48	13 - 25	54	21	31	13
PA02	81 - 98	25 - 35	12 - 19	54	21	31	13
PA03	95 - 122	31 - 46	14 - 22	54	21	32	13
PA04	90 - 122	30 - 46	14 - 22	54	21	32	13
PA05	90 - 109	30 - 49	14 - 24	54 - 55	21	32	13
PA06	91 - 125	30 - 50	14 - 23	54	21	32	13
PA07	101 - 138	39 - 63	21 - 33	56 - 57	22	33	13 - 14

ASR	NO_2 Concentration (μg/m³)			RSP Concentration (μg/m³)		FSP Concentration (μg/m³)	
	19th Highest Hourly Average	10th Highest Daily Average	Annual Average	10th Highest Daily Average	Annual Average	19th Highest Daily Average	Annual Average
AQOs	**200**	**120**	**40**	**75**	**30**	**37.5**	**15**
PA08	72 - 107	23 - 40	11 - 21	54	21	31	13
PA09	110	44	22	54	21	31	13
PA10	119	50	26	54	21	32	13
PA11	120	52	27	54	21	32	13
PA12	72 - 101	23 - 38	11 - 21	54	21	31	13
PA13	72 - 100	23 - 33	11 - 14	54	21	31	13
PA14	76 - 116	24 - 43	12 - 23	56	21 - 22	33	13 - 14
PA15	76 - 89	24 - 28	12 - 15	56	21	33	13
PA16	72 - 87	23 - 29	11 - 13	54	21	31	13

3.7.10 According to the discrete results, the most affected levels are observed at 1.5 mAG. The contour plots for the AQO pollutants at 1.5 mAG are presented in **60722948/A10/FIGURE 3.5a** to **3.11b**. Based on these contour plots, there are no AQO exceedance zones within 500 the assessment area, except for the 19th highest hourly NO_2 and the annual NO_2. As illustrated in **60722948/A10/FIGURE 3.9a-b** and **3.11a-b**, the exceedance zones are located in the top openings of the proposed shallow underpass and on the open roads within the Loop, and there are no planned or existing ASRs within these exceedance zones. As such, no adverse air quality impact is anticipated during the operation phase.

3.8 Mitigation Measures

Construction Phase

3.8.1 To minimise the exhaust emission from NRMMs during the construction phase, DEVB TC(W) No. 1/2015 (Emissions Control of NRMM in Capital Works Contracts of Public Works) shall be followed, and the following measures should be applied as far as practicable:

- Connect construction plant and equipment to main electricity supply and avoid use of diesel generators and diesel-powered equipment;
- Avoid exempted NRMMs; and
- Deploy electrified NRMMs.

3.8.2 In addition, air quality mitigation measures stipulated in the Air Pollution Control (Construction Dust) Regulation and good site practices listed below should be carried out to further minimise construction air quality impact.

- Regular watering to reduce dust emissions from exposed site surfaces and unpaved roads, particularly during dry weather.

- The construction will be carried out in sections or phases to minimise the impact on air quality for nearby air-sensitive receivers (ASRs).

- Frequent watering for particularly dusty construction areas and areas close to ASRs.

- Provide side enclosure and covering of any aggregate or dusty material storage piles to reduce emissions. Where this is not practicable owing to frequent usage, watering shall be applied to aggregate fines.

- Open stockpiles shall be avoided or covered. Where possible, prevent placing dusty material storage piles near ASRs.

- Tarpaulin covering of all dusty vehicle loads transported to, from and between site locations.

- The engine of the PMEs during idling shall be switched off.

- Provision of not less than 3m high hoarding from ground level along site boundary where adjoins a road, streets or other accessible to the public except for a site entrance or exit. Not less than 3.5m high hoarding should be used at the site boundary close to the ASRs with close proximity (e.g. A02, A12, A21).

- Where possible, routing of vehicles and positioning of construction plant should be at the maximum possible distance from ASRs.

- Locate all the dusty activities away from any nearby ASRs as far as practicable.

- Close liaison with the contractors of the concurrent projects to avoid overlapping of heavy/dusty construction works as far as practicable.

- All malodorous materials shall be placed as far as possible from any ASRs.

- The stockpiled malodorous materials shall be covered entirely by plastic tarpaulin sheets.

- The malodorous materials shall be removed from site as soon as possible and shall not be stockpiled overnight at the site.

- Loading of the malodorous materials onto the dump trucks shall be controlled to avoid spillage.

- Limiting the scale of excavation works at a time.

- Containing the odorous excavated materials in airtight/watertight containers on-site.

- A continuous dust monitoring programme will be implemented during the construction stage to enforce controls and modify method of work if dusty conditions arise and to ensure all nearby ASRs will not be subject to adverse construction air quality impact.

3.8.3 Guidelines stipulated in EPD's *Recommended Pollution Control Clauses for Construction Contracts* should be incorporated in the contract document to abate construction air quality impacts. These clauses include:

- The Contractor shall observe and comply with the Air Pollution Control Ordinance and its subsidiary regulations, particularly the Air Pollution Control (Open Burning) Regulation, Air Pollution Control (Construction Dust) Regulation, Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation, Air Pollution Control (Fuel Restriction) Regulations and Air Pollution Control (Smoke) Regulations;

- In addition to the statutory requirements of the Regulations, the Contractor of the public works contracts shall also observe the requirements as set out in the government circulars, including DEVB's TC No, 13/2020 (Timely Application of Temporary Electricity and Water Supply for Public Works Contracts and Wider Use of Electric Vehicles in Public Works Contracts) and DEVB's TC No. 1/2015 (Emissions Control of NRMM in Capital Works Contracts of Public Works);

- The Contractor shall undertake at all times to prevent dust nuisance and smoke as a result of his activities, and minimise the emission of air pollutants from construction plant and equipment;

- The Contractor shall ensure that there will be adequate water supply/storage for dust suppression;

- The Contractor shall devise, arrange methods of working and carrying out the works in such a manner so as to minimise dust impacts on the surrounding environment, and shall provide experienced personnel with suitable training to ensure that these methods are implemented;

- For better smoke control, the Contractor shall not use diesel hammer for percussive piling; and

- Before the commencement of any work, the Engineer may require the methods of working, plant, equipment and air pollution control system to be used on the site to be made available for inspection and approval to ensure that they are suitable for the project.

3.8.4 In order to help reduce carbon emission and pollution, timely application of temporary electricity and water supply should be made and electric vehicles should be adopted as far as practicable in accordance with *DEVB TC(W) No. 13/2020 – Timely Application of Temporary Electricity and Water Supply for Public Works Contracts* and *Wider Use of Electric Vehicles in Public Works Contracts*.

Operation Phase

3.8.5 No adverse air quality impact during the operation of the Project is anticipated. Mitigation measures are thus considered not necessary during the operation phase.

3.9 Evaluation of Residual Impacts

Construction Phase

3.9.1 With the implementation of the mitigation measures as stipulated in the Air Pollution Control (Construction Dust) Regulation together with the recommended air quality control measures and good site practices on the work sites, no residual impact would be expected from construction of the Project.

Operation Phase

3.9.2 No residual impact is expected during the operation phase of the Project.

3.10 Environmental Monitoring and Audit

Construction Phase

3.10.1 EM&A for potential air quality impacts should be conducted during construction phase so as to check compliance with the legislative requirements. Details of the monitoring and audit programme are contained in a stand-alone EM&A Manual.

3.10.2 Regular site audit for potential air quality impact, continuous dust monitoring is recommended to be conducted during the entire construction phase of the Project so as to ensure the air quality mitigation measures and the mitigation measures stipulated in Air Pollution Control (Construction Dust) Regulation are implemented in order and to ensure no adverse air quality impact at the ASRs.

Operation Phase

3.10.3 No residual air quality impact arising from the Project is anticipated during the operation of the Project. Therefore, the EM&A works for the operation phase are considered unnecessary.

3.11 Conclusion

Construction Phase

3.11.1 The potential air quality impacts which may arise from the construction works of the Project include site clearance and site formation works, slope works, construction of the underpass/viaduct, at-grade road sections, and wind erosion of exposed work area. Additionally, there may be emissions of PM and gaseous emission from the use of NRMM and construction trucks. With the implementation of mitigation measures specified in the Air Pollution Control (Construction Dust) Regulation together with the recommended air quality mitigation measures, good site practices, and EM&A programme, no adverse air quality impact at ASRs is anticipated due to the construction activities of the Project.

Operation Phase

3.11.2 Cumulative air quality impact arising from vehicular emission from the existing and planned open roads, emission associated with the existing minibus termini and the planned PTIs, industrial and portal emissions within 500m assessment area, as well as background concentrations, has been assessed for the operation phase of the Project. The results concluded that the predicted cumulative air pollutant concentrations on the

identified ASRs comply with the respective AQOs during the operation phase. As such, adverse air quality impact due to the operation of the Project is not anticipated.

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